MILITARY SPECIFICATION

INSULATION TAPE, ELECTRICAL, SEMI-CURED THERMOSETTING RESIN TREATED GLASS, ARMATURE BANDING, NAVAL SHIPBOARD

1. SCOPE

1.1 This specification covers semi-cured thermosetting resin treated, unidirectional glass electrical insulating tape intended for banding armatures of rotating electrical machines for shipboard use. Other applications may be banding of wound rotor coils, securing stator coils, surge rings, commutator string bands.

1.2 Classification - Glass banding tape shall be of the following types, as specified (see 6.2):

Type B - General purpose, 130°C. for continuous operation.
Type H - High temperature, 180°C. for continuous operation.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

FEDERAL
PPP-B-636 - Box, Fiberboard.

MILITARY
MIL-Y-1140 - Yarn, Cord, Sleeving, Cloth and Tape, Glass.

STANDARD

MILITARY
MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
D580 - Tests and Tolerances for Woven Glass Tapes.
D638 - Test for Tensile Properties of Plastics.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

FSC 5970

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[first article and the production items shall be identical and in

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, One Park Avenue at 33rd Street, New York, N.Y., 10016.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3.1 Preproduction.- Glass banding tape furnished under this specification shall be a product for which reports of preproduction inspection (see 4.2) have been submitted to the Bureau of Ships and have been found acceptable.

3.2 Construction.- The insulating tape shall consist of non-woven parallel ECG 150-2/2 glass yarns[1/] in accordance with MIL-Y-1140 and impregnated and bonded with a thermosetting resin. The tape shall be furnished in the semi-cured ("B" stage) state and shall be flat, dry and non-sticky. The surface may be slightly tacky but shall unwind freely from the spool without transfer of resin or glass strands. The thermosetting resin shall be such that the treated tape may be completely cured ("C" stage) after baking for 3 to 5 hours after the equipment reaches a temperature of 150°C. The tape shall consist of 80 ends $^{+0}_{-1}$ per inch width.

3.3 The tape shall be put up in rolls over a 3-1/4 inch diameter core. The tape shall be furnished in lengths of 200 yards per roll with no splices per roll and with no layer separators. Alternate put-ups may be supplied if satisfactory to the procuring activity. The number of rolls shall be as specified (see 6.2).

3.4 Thickness and widths.- The tape shall be furnished in the following thicknesses and widths, as specified (see 6.2):

Nominal thickness, inches
$0.014 \; ^{+\;0.004}_{-\;\;.002}$

Nominal width, inches
3/8 ± 3/32
1/2 ± 1/8
3/4 ± 1/8
5/8 ± 1/8
1 ± 1/8

3.5 Property values.- The insulating tape shall conform to the property values shown in table II (quality conformance tests) and table III (preproduction tests).

3.6 Instruction for storage.- The insulating tape shall have a minimum shelf life of 6 months at 20°C. (68°F.). Except for short shipping periods, the tape must be stored at 4°C. (40°F.). Lower temperatures have no deteriorating effects. An instruction sheet or label shall be included with each roll of tape giving complete storage instructions.

3.7 Workmanship.- The insulating tape shall be manufactured in a careful and workmanlike manner.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to

[1/] Other glass yarns with special finishes may be used if satisfactory to the Bureau of Ships.

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the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction inspection.- Preproduction inspection will be performed by the supplier prior to production. The tests shall consist of the hot banding test specified in 4.6.3 and the weight loss test specified in 4.6.4. A report of the test shall be submitted in duplicate to the Bureau of Ships. The initial test shall suffice for all future production provided no change is made in the tape as tested.

4.3 Sampling for quality conformance inspection.-

4.3.1 Lot.- All tape of the same composition, type and size manufactured under essentially the same conditions within a period of 15 days and offered for delivery at the same time shall be considered a lot for purposes of acceptance inspection.

4.3.2 Sampling for visual and dimensional examination.- Samples of tape shall be selected from each lot for examination of visual and dimensional characteristics in accordance with table I. The lot size and sample size shall be stated in terms of rolls.

Table I - Sampling for visual and dimensional examination.

Lot size number of rolls	Sample size rolls to be examined	Acceptance number (defective)	Rejection number (defective)
2 to 8	2	0	1
9 to 15	3	0	1
16 to 25	5	0	1
26 to 40	7	0	1
41 to 65	10	0	1
66 to 110	15	0	1
111 to 180	20	1	2

4.3.3 Sampling for quality conformance tests.- One fifth of the rolls selected for the visual or dimensional examination shall be subjected to the quality conformance tests specified in table II; however, the smallest number of rolls selected for any sample shall be one. If the contract calls for delivery of two rolls or less, only one roll shall be selected.

4.4 Quality conformance inspection.-

4.4.1 Visual and dimensional examination.- Each of the samples selected in accordance with 4.3.2 shall be visually and dimensionally examined to verify compliance with the requirements of 3.2, 3.3, 3.4, 3.5, 3.6 and 3.7. Any sample roll having one or more dimensional defects shall be considered defective. If the number of defective rolls exceeds the acceptance number for that sample, this shall be cause for rejection of the lot represented by the sample.

4.4.2 Quality conformance tests.- Quality conformance tests shall be conducted on samples of material selected in accordance with 4.3.3. Results shall be averaged for each sample roll to determine conformance with this specification. If the average for any sample roll fails to conform with any requirement, this shall be cause for rejection of the lot.

4.5 Conditioning.- Materials shall be conditioned before being tested, as specified in table II or table III.

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Table II - Quality conformance tests for property values of banding tape.

Property to be tested	Test paragraph	Number of specimen for each type	Test per specimen	Conditioning (see 4.5)	Unit of value	Property value	
						Type B	Type H
General examination 1/	4.6.1	see Table I	-	-	-	-	-
Thickness	4.6.1.2.1	-	10	A	inches	as required	
Width	4.6.1.2.2	-	5	A	inches	as required	
Resin content	4.6.1.3.1	-	5	A	percent, min.	25	25
Ends	4.6.1.3.2	-	5	A	min., per inch width	79	79
Bonding and curing	4.6.2	see 4.3.3	3	A	-	observation	
Tensile strength 2/	4.6.3	-	-	-	-	-	-
Cold Breaking load (cold band)	4.6.3	3	1	A	average, pounds	17,000	20,000
Hot Breaking load (hot band)	4.6.3	3	1	E168/130	average, pounds	13,000	18,000

1/ Materials, dimensions, and workmanship.
2/ Calculate from breaking load values and determine psi for information only.

Table III - Preproduction tests for property values of banding tape.

Property to be tested	Test paragraph	Number of specimen for each type	Test per specimen	Conditioning (see 4.5.)	Unit of value	Property value	
						Type B	Type H
Tensile strength 1/	4.6.3	-	-	-	-	-	-
Hot Breaking load (hot band)	4.6.3	3	1	E400/200	average, pounds	10,000	15,000
Weight loss	4.6.4	3	1	E400/200	average percent	17	10

1/ Calculate from breaking load values and determine psi for information only.

4.5.1 Equipment. - Conditioning chambers and baths shall be such as to maintain the specified temperature plus or minus 2°C. and the specified relative humidity plus or minus 2 percent.

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4.5.2 Designation.- Conditioning procedures shall be designated as follows:

(a) A capital letter indicating the general conditions of the specimen; that is humidity, immersion, and temperature conditioning.
(b) A number indicating in hours the duration of the conditioning.
(c) A number indicating in degrees centigrade the conditioning temperature.
(d) A number indicating relative humidity whenever relative humidity is controlled. (Relative humidity obtained over calcium chloride shall be taken as zero.)

The numbers shall be separated from each other by slant marks, and from the capital letter by a dash.

4.5.3 Nomenclature.- The following letters shall be used to indicate the respective general conditioning:

Condition A - As received; no special conditioning
Condition E - The specimen shall be conditioned for the time and temperature specified.

Examples:

E - 6/130 - 6 hours, 130°C.
E - 168/180 - 168 hours, 180°C.

4.6 Methods of tests.- The methods of test and procedures shall be as specified in 4.6.1 to 4.6.4, inclusive. Unless otherwise specified herein, all tests shall be conducted under room conditions (23°C., 50 percent relative humidity).

4.6.1 General examination.-

4.6.1.1 The insulation shall be examined to verify that the dimensions, materials, labeling and workmanship are in strict conformance with this specification. (See 3.2 through 3.7.)

4.6.1.2 Methods of measurement of dimensions.-

4.6.1.2.1 Thickness.- The thickness of the tape shall be determined in accordance with the method specified in ASTM D580.

4.6.1.2.2 Width.- The width shall be determined by the use of a standard steel scale graduated to read in 1/16 inch. The tape shall be pressed flat on a smooth surface at the time of representative points on the material at least 1 yard apart and the maximum, minimum and average valves recorded.

4.6.1.3 Materials.-

4.6.1.3.1 Resin content.- The resin content shall be determined in accordance with the organic content method of ASTM D580. When constant weight is attained calculate the loss in weight.

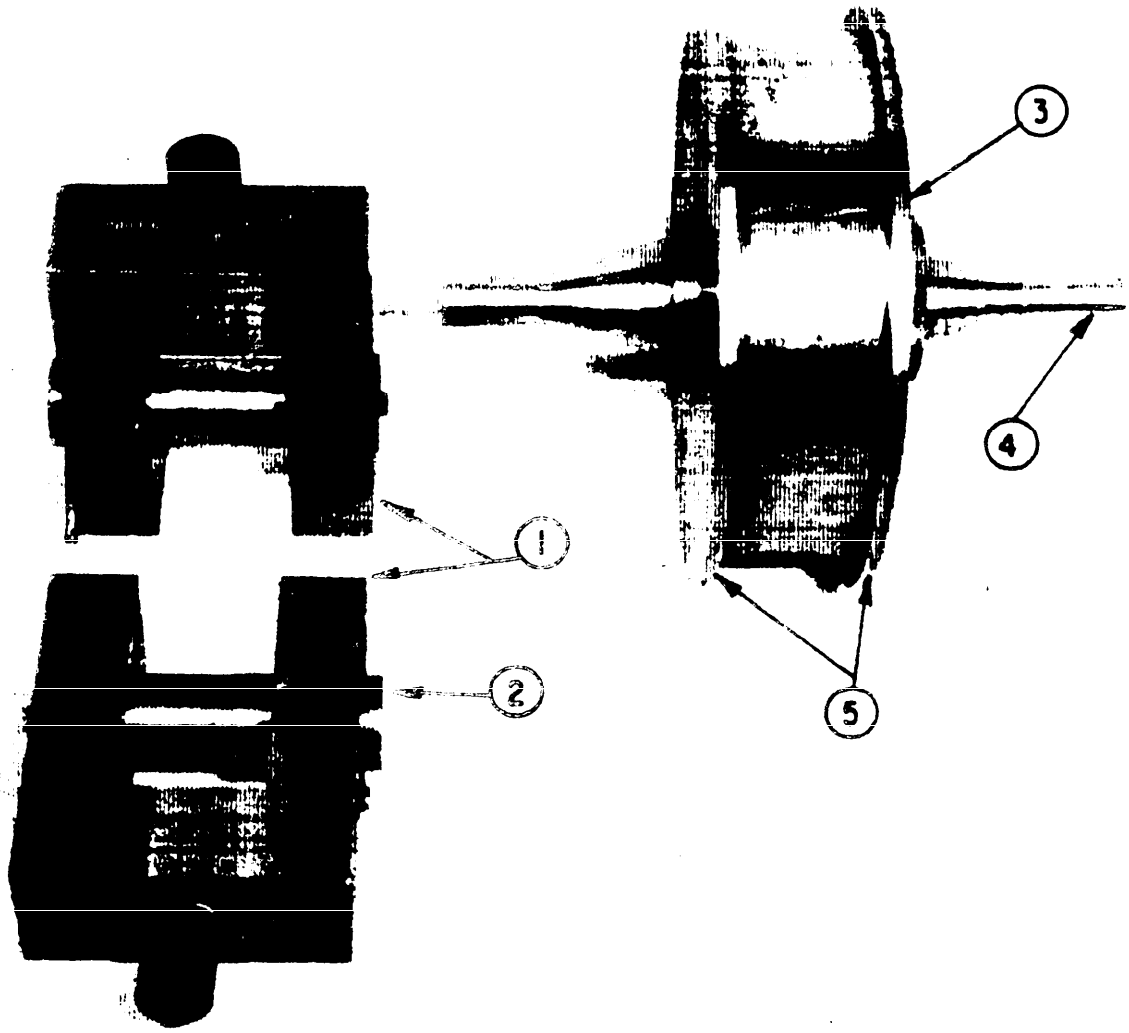
4.6.1.3.2 Number of ends.- The glass construction shall be examined to determine the number of ends of glass in the tape. The samples resulting from the resin content test of 4.6.1.3.1 may be examined for actual ends content or new samples may be examined by use of a magnifier of approximately 200 power. An end is defined as a twisted and plied yarn. ECG 150 2/2 glass consists of 2 strands twisted and 2 yarns plied; therefore with this construction 1 end = 2 yarns = 4 strands.

4.6.2 Bonding and curing.- The tape shall bond to itself after heating under pressure for 20 minutes at 150°C. and shall be fully cured after heating 5 hours at 150°C.

4.6.3 Tensile strength.- The tensile strength shall be determined on glass bands formed and tested using the split mandrel test jig.

4.6.3.1 Preparation of samples.- Bands shall be prepared by winding the glass tape under 500 pounds minimum tension, per inch of width, flat and free from buckles, on to a split mandrel in accordance with figures 1 and 2. Eight turns shall be wound per band (8 layers of glass on each side) and each band shall be one inch wide. Two one-inch bands may be wound on the mandrel by using a center spacer ring 1/2-inch wide and 1/4-inch thick. Terminal joints in the bands may be made by applying a hot clean soldering

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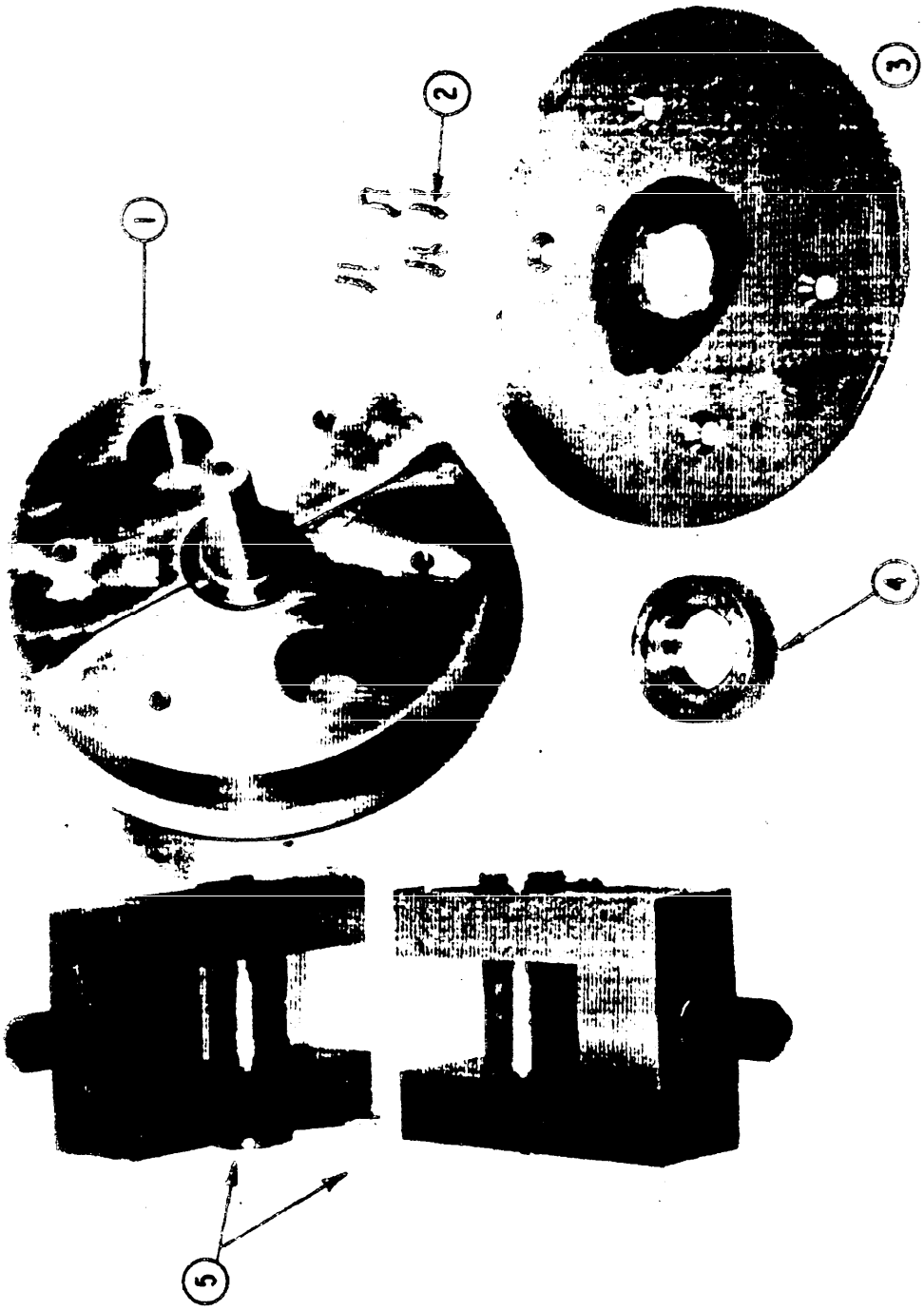


SH 8553

1 - Clevises
2 - Removable Pin
3 - Split Mandrel
4 - Shaft for Lathe Mounting
5 - Side Plates - Used When Winding and Removed for Tensile Tests

Figure 1 - Split mandrel and clevises used for forming and testing tensile specimens.

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1 - Mandrel Halves Held in Position with Remaining Side Plate
2 - Side Plate Positioning Screws
3 - Removed Side Plate
4 - Shaft Fastening and Alignment Nut
5 - Clevises for Tensile Machine Mounting

Figure 2 - Split mandrel and clevises used for forming and testing tensile specimens showing side plate removed.

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iron to the material for approximately 15 seconds upon each part of a 3-inch lap. Start and termination of the bands shall be made 90 degrees from the horizontal split of the mandrel. After winding and terminating, the bands shall be cured on the mandrel for 5 hours at 150°C. Bands prepared in accordance with the above procedure shall be known as "cold bands". A release agent may be used to allow the cured bands to be removed from the mandrel. Hot banding samples will also be required. The procedure for sample preparation shall be the same as above except the tape shall be heated prior to winding on the mandrel by using a heat lamp or other means sufficient to cause the resin to flow freely as the tape is wound on the fixture. The ends shall be securely tied during the curing (on the mandrel) cycle to prevent any slippage.

4.6.3.2 Tensile testing.- The tensile test shall be conducted in accordance with the procedure described in ASTM D638 using the glass band specimens, and a tensile test machine of a maximum of 30,000 pounds scale reading, and operating at a speed of 0.05 inches per minute. Where hot breaking load is required to be determined, the split ring test fixture and the test band shall be preheated to 130°C and the band strength or breaking load determined at this temperature ± 5°C.

4.6.3.3 Data and calculations.-

4.6.3.3.1 Breaking load.- The breaking load in pounds shall be recorded for each band tested and for the average breaking load for all specimens in the sample.

4.6.3.3.2 Tensile strength.- The tensile strength shall be calculated by dividing the average breaking load in pounds by the average cross sectional area of the glass content of the band 1/ in square inches. Express the result in pounds per square inch.

4.6.4 Weight loss.- The weight loss shall be determined in accordance with the organic content method of ASTM D580 except the weight B shall be determined after oven heating 400 hours at 200.C±5°C (392°F±9°F). The weight loss shall be calculated on the basis of the composite weight (resin and glass) of completed bands. Surplus resin squeezed out during processing shall be removed prior to initial weight measurement.

4.7 Inspection of preparation for delivery.- Packaging, packing, and marking shall be inspected to assure compliance with section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.- Banding tape shall be furnished in 250 yard continuous lengths on 3-inch ID bushings. Each roll shall be sealed in a moistureproof bag or film.

5.1.2 Level C.- Banding tape shall be packaged in accordance with the supplier's commercial practice.

5.2 Packing.- Packing shall be level A or C, as specified (see 6.2).

5.2.1 Level A. Rolls of tape, packaged as specified in 5.1.1, shall be packed in fiberboard boxes, class weather-resistant, in accordance with PPP-B-636.

5.2.2. Level C.- Insulation shall be packed in a manner which will insure carrier acceptance and safe delivery at destination. Shipping containers or method of packing shall comply with the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation.

1/ The glass area of all individual filaments (ECG-150 2/2 glass) in the glass tape shall be considered as 0.0074 square inches per inch width of tape per turn. This is based on 80 ends (of glass) per inch width and a filament diameter of 0.00038 inches. Each band, therefore, shall be considered as having a total glass area of 0.1184 square inches (0.0074 by 8 layers by 2 sides/band = 0.1184).

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5.3 Marking. - In addition to any special marking required by the contract or order or herein, interior and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.1 Special marking. - In addition to the marking requirements specified above, each unit and intermediate package shall be marked with the month and year of manufacture. Exterior shipping container shall be marked on two opposite sides with the following:

"Store at 40°F or lower temperature for maximum shelf life. Do not remove contents until ready for use".

6. NOTES

6.1 Intended use. - This tape is intended for banding armatures and other items of electrical equipment where a strong, non-metallic restraining material is needed to prevent coil movement under centrifugal forces and electrical stresses. Instructions for applying these materials to armatures and other items of electrical equipment is contained in the appendix to this specification.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number and date of this specificaion.
(b) Type required (see 1.2).
(c) Number of rolls (see 3.3),
(d) Width required (see 3.4),
(f) Applicable level of packaging and packing required (see 5.1 and 5.2).

6.3 Preproduction approval. - With respect to preproduction inspection (see 4.2) and approval (see 3.1), no tape may be offered for delivery until this requirement is met. As the testing requires a minimum of 400 hours aging time suppliers are urged to obtain preproduction approval prior to acceptance of any contract or order to insure delivery of their products without delay.

Preparing activity:
Navy - SH
(Project 5970-N082Sh)

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APPENDIX I

PROCEDURE FOR GLASS BANDING OF ELECTRICAL EQUIPMENT

10. SCOPE

10.1 Scope.- This appendix covers the procedure for banding dc armatures and ac wound rotors. The procedure may also be followed in securing surge rings and similar applications and for applying commutator string bands.

20. APPLICABLE DOCUMENTS

20.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-997 - Plastic-Material, Laminated, Thermosetting, Electrical-Insulating, Sheets, Glass Cloth.
MIL-P-15037 - Plastic Sheet, Laminated, Thermosetting, Glass-Cloth, Melamine-Resin.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

30. REQUIREMENTS

30.1 Glass banding material.- Glass tape for banding purposes is made of non-woven fibrous glass yarns laid parallel and impregnated with various thermosetting resins. The strength of the tape is the result of the many parallel fibrous glass filaments so oriented that all filaments share the tensile load and the glass-to-resin ratio is high. Type B banding tape shall be used for class A (105°C) and class B (130°C) insulation systems. Type H banding tape shall be used for class F (155°C) and class H (180°C) insulation systems.

30.1.1 Advantages over steel wire bands.- Glass bands have the following advantages over steel wire bands when used for holding coils and windings in place:

(a) Eliminates the need for insulation to be provided underneath the band.
(b) As most glass bands have good arc resistance, flashover failures may be reduced.
(c) Eliminates wire lashing and whipping associated with breakage of steel wire bands.
(d) Overheating up to 260°C. (500°F.) for short periods of time may be tolerated; whereas, solder may melt on steel wire bands.
(e) Application of glass bands to armatures is relatively easy and safe.

30.1.2 Banding tape with cross threads.- Some commercially available tapes are made with cross threads of glass to prevent separation of unidirectional fibers. These cross threads result in some loss of strength of bands. Type B and type H tapes do not contain cross threads.

30.1.3 Storage.- Banding tape is normally refrigerated (40°F.) to provide a useful storage life of 6 months. Exposure to temperatures above 20°C. (68°F.) for extended periods may make the tape useless. When its shelf life has passed, the resin has cured and it is no longer soft and pliable. Close watch of ordering and use schedules should be kept.

30.1.4 Tacky tape.- The banding tape is packaged in a semi-cured state and is usually relatively dry and non-sticky. Some tapes may become tacky and difficult to run through the tension device if they become too warm. Return of such material to a refrigerator for a short period of time will cool the tape and assist in reducing the tacky condition.

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30.1.5 Curing. - The curing cycle of banding materials is from 4 to 5 hours at a temperature of 135° to 150°C. (275° to 302°F.). When in doubt, 5 hours at 150°C. (302°F.) should insure a proper cure. A higher temperature will accelerate the cure. Curing of the resins into satisfactory bands will not occur at temperatures below 120°C. (248°F.) regardless of the length of exposure time. When the resin cures, it becomes a stiff, hard, springy, homogeneous mass binding the glass fibers and anchoring the band in place.

30.1.5.1 Phenolic varnishes (and air in the case of some polyester) may slow down the cure of some resins, although they usually will not prevent cure. Where the possibility exists that this may cause such trouble, a layer of very thin plastic film (polyester) covering the band during the curing cycle will prevent difficulties. When such a film is used on the band, the release (nonadhering) side of the film should be placed next to the band to keep the film from sticking to the band after curing the resin. After curing the band, the covering is untied and removed.

30.1.6 Precautions. - When using resin-impregnated glass tape for banding, several precautions are necessary to insure a completely satisfactory job.

(a) Age. - Be sure the material used for the band is not past its useful shelf life.
(b) Use schedule. - Tape should be stored at 4°C. (40°F.) for a shelf life of no more than 6 months (use old stock first).
(c) Critical temperature. - When baking the armature to cure the resin-impregnated glass band, check the oven to be sure the correct temperature is being used. A bad thermometer can mean uncured bands.
(d) Inhibiting agents. - Take precautions against inhibiting cure by exposure to phenolic varnish or other materials. Some polyester resins are air inhibited and should be covered during cure cycle.
(e) Limits. - Never bring the band as far as the core, since this would prevent varnish treatment from getting to the coil insulation under the band.

30.1.7 Magnetic steel-wire bands. WARNING! Do not replace magnetic steel-wire bands with glass bands because the magnetic steel-wire band is usually part of the magnetic circuit, carrying part of the commutating pole flux. A change to a glass band is apt to disturb the flux configuration in the commutating zone with consequent commutation problems. This could involve changing the commutating air gaps and field strength to restore the commutation to an acceptable level. Magnetic steel-wire bands can be readily detected with a magnet. Some Bureau of Ships Instruction Manuals make reference to the type of bands in the text or on the included drawings. Occasionally armatures are observed having had magnetic steel-wire bands installed in error at some previous date. When any question arises regarding the proper band replacement contact the Bureau of Ships.

30.2 Preparation of banding. -

30.2.1 Step one: Quantity needed. - Determine the amount of glass band.

30.2.1.1 When banding an armature that was formerly banded with a steel-wire band, use a glass band with twice the cross-section area of the original steel-wire band. If the width of the glass band is to be the same width as the steel band, the total thickness of the glass band must be twice the thickness of the steel band it replaces. The glass band can be at least as wide as the insulation under the steel band; therefore, by making it wider than the original steel band, its thickness can be reduced proportionally. Since no insulation is needed under the glass band, the total outside diameter over the top of the glass band generally will be not much greater than that over the top of the steel band.

Example. If the drawing or inspection of the band removed from the armature shows a double steel-wire band of 0.072-inch-diameter wire and a bandwidth of 3 inches, the equivalent area of the glass band will be twice that of the steel band, or two times (0.072 inch thick by 3 inches wide by 2 layers) 0.864 square inch.

30.2.1.2 Assuming that the steel band insulation pad was 3-3/4 inches wide, the glass band can be 3-3/4 inches wide. Then, the minimum thickness of the glass band will be 0.864 square inch divided by 3.75 inches wide or 0.230 inch thick. It can be assumed that each layer of band (cured) will be 0.010 inch thick, so there will be about 23 layers of banding tape. When the top of the band is to be machined for any reason, add an additional 1/32 inch, or three turns, to be machined off.

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30.2.2 Step two: Tension. - Determine the pull-down tension, also called winding tension, of the glass tape. It should be the same as the recommended steel-wire banding tension. In replacing two layers of steel wire, the average tension used for the outer and inner layers should be used for glass tape tension. This tension will pull the coils into their final position and will set up the glass, initially, so that the centrifugal force of the windings will not result in movement during operation of the equipment.

Caution: Care must be used with small armatures so that excessive tension will not be used with resultant deforming of the coils being held in place.

30.2.2.1 Maximum tension of 600 pounds per inch of tape width should be used (i.e., 300 pounds maximum for 1/2-inch-wide tape).

30.2.3 Step three: Boundaries. - Locate edges or boundaries of the glass bands on the armature. The edge of the band should not butt against the armature core, since this may prevent penetration of the varnish into the coil area under the band. On small armatures where less than 200-pound tension will be used on the banding tape, the boundary should be marked by a band of ordinary adhesive tape or other convenient means. When more than 200-pound tension will be used on the banding tape, side restraints will be needed. Since the banding tapes do not have lateral strength, side restraints are needed to prevent side slippage during application and cure of the resin in bands under high tension. The side restraints should be at least as high as the band. Where economical, reusable fixtures, made of a material that is nonadhesive to the resin in the banding material or treated with a release (nonadhering) coating, may be used. A satisfactory disposable edge restraint may be made from a strip of Fullerboard or similar material wrapped around the armature and held in place by a few turns of banding wire or a shipping strap. Most convenient for repair work are edge-restraint strips consisting of glass and absestos cordage, unimpregnated, and with a tail to be held in place by the band. They are commercially available in two sizes. The smaller, for bands up to 1/8 inch thick, consists of a 5/32-inch welt and 1-inch tail. The larger, for bands greater than 1/8 inch thick, has a 5/16-inch welt and a 3-inch tail. This type of restraint becomes a part of the band and is not removed after the band is cured.

30.2.4 Step four: Covering strips. - Prepare covering strips to cover large opening between windings in the band area. This prevents the lower turns of glass tape from being forced into the gaps by the upper layers of the band. Material for this covering may be glass melamine (MIL-P-15037, type GME) or glass silicone laminate (MIL-P-997, type GSG). It should be 0.032 inch thick and cut into strips wide enough to cover the gaps and approximately 9 to 12 inches long (as required by the armature size) with the edges lapped about 1/2 inch to permit "shingling" as the band tension snugs the coils down against the coil supports.

30.2.5 Step five. Core filler strips. - Where the coils are below the core surface, filler strips of glass melamine, slightly less than slot width between the coils and band are needed to insure that full pressure is maintained to pull the coils down to the bottom of the slots. Where filler strips are required, the edges should be smoothly rounded to prevent breaking of the glass yarns in the band.

30.2.6 Step six. Temporary bands. - Apply temporary bands in the same manner as for steel banding. Banding tension should not be relaxed at any time during application of bands.

30.2.7 Step seven: Balance. - Balance the armature before winding the tape in place so that the weights can be incorporated in the band, if this method of attaching balancing weights is required.

30.3 Tension Devices. - The tension device for applying resin-impregnated glass bands to armatures must have flat, rotating pulleys and a tension drum and be capable of creating a steady, uniform tension of up to 600 pounds without damage to the tape.[1/] The device should feed the tape straight to the banding surface without twists or turns. The tape should be allowed to feed into the tension device by unwinding freely from a spool on a shaft that is well clear of the tension device.

[1/] The wire banding tension devices with grooved pulleys or drag blocks cannot be used with glass tapes.

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30.3.1 Devices for adapting standard banding machines or a lathe to glass tape banding are available from several suppliers as follows:

(a) Westinghouse Electric Corp. (complete wire tension units and adapter for glass tape)
543 North Lang Avenue
Homewood, Pa.

(b) Potter & Rayfield, Inc. (Type FA unit accommodates either wire or glass type)
1570 Northside Drive
Atlanta, Georgia

(c) Klein Armature Works (For glass tape only)
1439 North Elm Street
Centralia, Illinois

(d) U. S. Navy Marine Engineering Laboratory
Annapolis, Maryland 21402

(For those shops desiring to construct their own tension device drawings, instructions and parts lists are available from U. S. Navy Marine Engineering Laboratory. Request copy of report 810199 dated 4 April 1961 direct from the U. S. Navy Marine Engineering Laboratory.)

30.4 Banding operation. - After preparations for banding are complete, including the installation of temporary bands, the banding operation can start. The armature should be preheated to a temperature of 100° to 135°C.(212° to 275°F.). It is then mounted in the banding machine or lathe that has been fitted with a tension device suitable for applying tension to a resin-impregnated glass banding material.[2/]

30.4.1 Steps. - Banding includes the following steps in the order of their occurrence.

(a) Secure in place the covering strips, if needed, to close any large openings in the armature winding in the band area. This can be done with a string or cord which can be removed as the band covers the strips.
(b) Tie a band of cotton tape or other suitable material around the core and attach one end of the glass tape from the tension device to the cotton tape. Then take one full turn around the armature to hold the glass tape.
(c) Adjust the tension device to the desired tension. Proceed to apply the band, starting at the core end of the band and moving to the outer edge of the band; then wind the tape back and forth for the required depth until the band is completed.
(d) Secure the last 4 to 6 inches of the end of the tape to the band with a hot, clean soldering iron. After the end is sealed, cut the tape.
(e) Repeat the above procedures at each band of the armature.
(f) Place the armature now with the resin-impregnated bands in place in an oven for curing at 150°C.(302°F.) for 5 hours, or as indicated by the tape manufacturer's literature.
(g) Remove the edge restraints (unless they are built in) and remaining temporary bands.

30.4.2 Postbanding treatment. - After the cure cycle is finished, it is desirable to remove any rough edges left on the band by light dressing with a file as the armature is turning. This can be done while machining the commutator or balancing the armature. The varnish treatments and other operations necessary to complete repair of the armature can now be completed and the armature reinstalled.

30.5 Balancing the armature. - All balancing should be done as far as possible without adding weight to the band. Where it is necessary, however, to balance by adding weights to the band, or to complete the balance by adding a small weight to the band, three methods are possible and, if properly carried out, will result in a satisfactory operation.

30.5.1 Early method. - The first method is acceptable provided the coils are in or near their final location as a result of temporary banding and will not be disturbed appreciably by final banding. In this method, final balancing is done before the glass bands are placed, and the balance weight (a piece of sheet lead, 1/16 inch thick) is cut and weighed by the balancer to suit and is located by marking the armature

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[2/] Hereafter the term "banding machine," alone, will be used to mean either.

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where the weight is to be applied. This weight is placed on a piece of flexible mica and, after one or two layers of the banding tape are applied, is held in place as marked and wound under succeeding layers.

30.5.2 Band turned down. - In the second method, the band is laid up about 1/32 inch thicker than necessary, and this additional thickness is turned off after the band is completely cured. This can be done at the time the commutator is turned. It leaves a smooth contour for the addition of balance weights. Balancing is done by placing weights at the proper points on the band to achieve balance. Sheet lead about 1/16-inch thick which has been copper plated can be secured to the band by an epoxy adhesive. A balancing weight so placed on an armature should be secured in place by several complete turns of a thermosetting glass adhesive tape around the armature.

30.5.3 Weighted epoxy. - A third method of balancing uses a lead weighted epoxy that can be applied to the glass band for balance. The lead weighted epoxy is available with a catalyst. Since the material mixed with the catalyst has a useful life of 15 to 30 minutes, mixing must be done only as immediately required. The compound is placed on the cleaned surface of the band where the weight is needed. A 250-watt infrared lamp or curved heat unit placed 12 inches away will harden the compound in 15 minutes sufficiently to permit further balancing. Air cure occurs in 1 to 2 hours.

40. QUALITY ASSURANCE PROVISIONS

40.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification: where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

40.2 Test after banding operations completed. - There are no special tests conducted on the bands after curing. They should be observed to see that they are hard, glossy, smooth, well filled with resin and should ring like a bell when struck. Tests on the finished electrical equipment should be as required by the applicable equipment specification.

50. PREPARATION FOR DELIVERY

50.1 This section is not applicable to this appendix.

60. NOTES

60.1 This section is not applicable to this appendix.

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